

vasic-digital tmux — Project Constitution

Field	Value
Revision	4
Created	2026-05-07
Last modified	2026-08-10
Status	active
Extends	constitution/Constitution.md — Helix Universal Constitution, pinned 7f738df
Status summary	R4 (2026-08-10): §105/§106 updated — no CPU/task/lifetime cap by default (matching the already-correct "liquid" memory model); root-cause fix for sessions/processes being killed despite ample host capacity (idle-recycler default-on kill + unconditional CPUQuota/TasksMax with no full opt-out). R3 (2026-05-21): full refactor to the HelixConstitution extends-template form. Universal clauses now inherited from the constitution/ submodule; this file holds only Project Articles §101–§109. No universal clause weakened — inheritance is verified by scripts/tests/test_constitution_inheritance.sh + its paired mutation.

This constitution **extends** the Helix Universal Constitution at [constitution/Constitution.md](#). Every clause there — the anti-bluff covenant (§11.4), absolute data safety (§9), the 60 % host-memory budget, the continuation-document invariant (§12.10), the no-guessing mandate (§11.4.6), commit/push discipline, credentials handling — applies to this repository **unconditionally**.

The Project Articles below **extend** those universal clauses with tmux-specific rules. They never weaken or override a universal clause. **When this file disagrees with the constitution submodule, the constitution wins.**

Inheritance is verified, not assumed. `scripts/tests/test_constitution_inheritance.sh` asserts the submodule is present at the pinned revision and that the §11.4 *End-user Quality Guarantee* anchor physically exists in `constitution/Constitution.md`. Its paired mutation — `constitution/`

```
meta_test_inheritance.sh, wired as the CM-CONSTITUTION-
INHERITANCE entry in scripts/tests/
meta_test_false_positive_proof.sh — deletes that anchor and
proves the gate then FAILs. The gate cannot itself be a bluff
gate.
```

Project Articles

§101 — Anti-bluff covenant binding (origin anchor)

This repository is one of the historical origins of the Helix anti-bluff covenant. The verbatim user mandate that the universal §11.4 codifies is preserved here for forensic continuity — it **MUST** remain in this project's Constitution regardless of inheritance:

```
"We had been in position that all tests do execute with
success and all Challenges as well, but in reality the most of
the features does not work and can't be used! This MUST
NOT be the case and execution of tests and Challenges
MUST guarantee the quality, the completion and full
usability by end users of the product!"
```

The bar for shipping is "users can use the feature," not "tests pass." Every PASS in this codebase — unit test OR HelixQA Challenge — **MUST** carry positive runtime evidence captured *during execution* that the feature works for the end user. Metadata-only PASS, configuration- only PASS, "absence-of-error" PASS, and grep-only PASS are critical defects regardless of how green the summary line is. **FAIL-bluffs** (a test that exits FAIL for a script-internal reason, not a product defect) are equally forbidden and **MUST** be fixed at the source layer.

The full covenant — recorded-evidence requirement, FAIL-bluff prohibition, no-guessing mandate, test-interrupt-on-discovery — is universal §11.4.1 through §11.4.6. This article binds this project to it and adds the project-specific articles below.

§102 — Operator-path test coverage (project rule, User mandate 2026-05-13)

Forensic anchor: Fixed.md A12 + A13. Tests 11 and 14 reported GREEN while their operator-facing equivalents (tmx new -s X) were broken — test 11 only exercised the explicit-socket path; test 14 hand-spawned systemd-run --user --scope units while the real tmx new workflow placed all sessions in one shared cgroup.

The mandate. Every gate test for a feature **MUST** exercise the SAME entry point an end-user would invoke in production. Tests that bypass the operator's wrapper / helper / install path and

reproduce its effects with hand-crafted equivalents DO NOT satisfy the universal §11.4.2 captured-evidence requirement. When the operator's path and the test's path diverge:

1. The test header MUST EXPLICITLY name the divergence (e.g. "test invokes tmux directly; operator path is tmx new").
2. A SEPARATE end-to-end test MUST close that divergence with captured evidence on the operator-facing entry point.

Operative test: for every test under scripts/tests/, ask "would an operator hit this code path in their normal workflow?" If no, that test is supplementary; the operator-path test must exist alongside it.

No grep-on-script-content alone. A grep on a script for a literal string is allowed AS A STATIC CHECK *in addition to* a runtime readback — never as the only assertion.

Layer-4 mutations MUST target the operator-path code. Mutations target scripts/tmx-vm / scripts/tmx.template / scripts/tmux.conf.template (the code that runs in production), not synthetic-test code.

Non-compliance is a release blocker.

§103 — Four-layer test coverage for tmux defects (User mandate 2026-05-06)

Every defect found, fixed, or introduced in this repo MUST land all four layers before the cycle closes:

- **Layer 1 — pre-build / static-source gate.** A check in scripts/verify.sh (or the relevant build script) that catches the defect class at source.
- **Layer 2 — runtime test.** A numbered test in scripts/tests/NN_*.sh producing positive runtime artifacts, anti-bluff per §101, operator-path per §102, topology-dispatched per §104.
- **Layer 3 — HelixQA Challenge.** An entry in scripts/challenges/tmux.yaml carrying captured-evidence semantics. A test without a paired Challenge is a §101 PASS-bluff.
- **Layer 4 — paired mutation.** A mutation in scripts/tests/meta_test_false_positive_proof.sh proving the Layer-2 test catches its own negation (mutation CAUGHT) and that the feature is restored afterwards (RESTORED).

Test-interrupt-on-discovery. The moment any defect is rediscovered, re-produced, or newly identified during a test cycle, the cycle STOPS. Fix at root cause → land all four layers → full rebuild → repeat the full cycle from the beginning.

§104 — Per-host-topology test dispatch for tmux (User mandate 2026-05-05)

Tests that depend on host topology (systemd version, cgroup v1 vs v2, distro-specific controller availability, presence of systemd-run --user, Linux vs Darwin) MUST detect topology at test entry and dispatch the topology-appropriate variant — recording the probe output as positive evidence of the SKIP rationale. A test running the wrong variant for the actual topology and PASSing is a §101 PASS-bluff. Silent degradation to a no-isolation path is forbidden.

scripts/tmx classifies the host as tmx-supported / tmx-degraded / tmx-unsupported via `_probe_topology()` — the canonical dispatch seam.

§105 — Per-session isolation architecture (default since 2026-05-13; no-limits-by-default since 2026-08-10)

Each `tmx new -s NAME` spawns its OWN tmux server on socket `tmx-NAME` inside its OWN OS-native isolation boundary. **The isolation boundary gives every session an independent crash/identity container (§14); it carries NO resource or lifetime cap unless the operator explicitly opts in.** Root-cause forensic anchor (2026-08-10, operator mandate): sessions and their processes were observed killed/wiped out despite running on powerful hardware with ample capacity. Systematic-debugging traced this to three unconditional-by-default caps (CPUQuota with no off switch, a hardcoded `TasksMax=4096` with no override knob at all, and — the dominant cause — the idle-timeout recycler tearing down ANY detached session after 900 s regardless of whether it was actively working). All three are now opt-in only, extending the memory model (§106) that was already correct:

- **Linux** — cgroup-v2 transient scope `tmx-NAME.scope` via `systemd-run --user --scope`, **fully elastic ("liquid") memory**: `MemoryMax=infinity` (a session is NEVER per-scope OOM-killed) plus a `MemoryMin=128M` floor so it is not reclaimed to death. `TMX_MEM` (or `TMX_MEM=auto` for the host-adaptive size) adds an OPT-IN *soft* `MemoryHigh` throttle (reclaim, never kill). **CPU: no CPUQuota property at all by default (unlimited)** — `TMX_CPU=auto` opts IN to the v1.0.37 host-adaptive value (`cores × 15%`), `TMX_CPU=<pct>` opts IN to an explicit cap. **Tasks: TasksMax=infinity by default** — `TMX_TASKS=auto` opts IN to the legacy fixed 4096 ceiling, `TMX_TASKS=<N>` opts IN to an explicit cap. `Delegate=yes`. Genuine exhaustion is handled system-wide by `systemd-oomd` + `zram` + the `user.slice` backstop (see the OOM-Protect project): a real runaway is stopped at TRUE exhaustion without freezing the box, while normal multi-session load never triggers a premature kill. Scopes remain independent cgroups (§14 crash isolation). An OPT-IN `TMX_SERVER_SPLIT=1` topology additionally isolates the tmux

SERVER's own small scope from the workload's cgroup so a bursting fleet can never CPU-starve the server's own keystroke/echo/timer handling (requires an explicit, splittable TMX_CPU to engage — see scripts/tmx-pane-shim.sh + the `_split_cpu_pcts` design comment in scripts/tmx.template).

- **macOS (Darwin)** — POSIX rlimit wrapper as session default-command: RLIMIT_CPU + RLIMIT_NPROC, kernel-enforced, **both unlimited by default** — `TMX_CPU_HARD_SECS=<secs> / TMX_PROC_MAX=<N> opt IN` to an explicit cap (was a fixed 24 h / 4096 default with no way to remove it). RLIMIT_AS (memory) is NOT enforced by XNU for unprivileged processes — documented honestly per §101, see docs/guide/README.md §5.6.
- **Both OSes** — the idle-timeout session recycler (scripts/tmx-recycler.sh) is OFF by default (`TMX_RECYCLE_IDLE_SECS=0`). `TMX_RECYCLE_IDLE_SECS=<secs> opt IN` to auto-teardown of a session after that many seconds with no client attached (state — `last_pwd/color/password` — always survives a recycle; only `tmx delete` clears it).

The session shell is the operator's host shell with full \$PATH. Plain-vanilla tmux UX with OS-native isolation as a safeguard, and no resource/lifetime ceiling unless the operator asks for one.

§106 — Resource-budget enforcement (project enforcement of universal §12.6) — CPU, memory, tasks, and lifetime all opt-in

Default is **fully elastic across every dimension**: sessions use all available CPU/RAM/zram/task-slots, run for as long as the operator wants, and are never per-scope OOM-killed, throttled, task-capped, or auto-torn-down by tmx itself. The universal 60 % host-memory budget is enforced system-wide at the `user.slice` level (OOM-Protect: `MemoryMax=95%`, `MemoryHigh=90%`) rather than per session, so genuine exhaustion is caught without capping individual sessions — "never kill prematurely, never stuck." Every cap is strictly OPT-IN, via environment variable (settable as durable shell-rc configuration or as a per-invocation param — both are "configuration" in this project's design):

Knob	Default	=auto opt-in	Explicit opt-in
TMX_MEM	unset → no soft throttle (hard cap always infinity)	host-adaptive max(MemTotal × 60% / 4, 2 GB) <i>soft</i> MemoryHigh	TMX_MEM=8G etc.
TMX_CPU	unset → no CPUQuota (unlimited)	host-adaptive cores × 15% (floor 200%,	TMX_CPU=400 (%)

Knob	Default	=auto opt-in	Explicit opt-in
		cap cores × 100%)	
TMX_TASKS	unset → TasksMax=infinity	legacy fixed 4096	TMX_TASKS=8192
TMX_RECYCLE_IDLE_SECS	unset/0 → never auto-recycled	— (no auto form)	TMX_RECYCLE_IDLE_SECS=90 etc.
TMX_CPU_HARD_SEC (Darwin)	unset → unlimited RLIMIT_CPU	—	TMX_CPU_HARD_SEC=3600
TMX_PROC_MAX (Darwin)	unset → unlimited RLIMIT_NPROC	—	TMX_PROC_MAX=4096

On macOS the memory cap is informational only regardless of TMX_MEM (XNU gap, §105).

§107 — Upstream tmux submodule pinned + immutable

tmux/ references tmux/tmux at tag 3.6a (commit cc117b5). It is a third-party upstream submodule: **never modified**, never advanced off the tag without an explicit, documented decision in Fixed.md + a governance-doc update in the same commit. Silent pin-drift is a §101 / universal §11.4.6 violation (forensic precedent: Fixed.md §3.8).

§108 — Native dual-OS build discipline

The hardened tmux binary is built natively per OS — Linux ELF and macOS Mach-O — from the same tmux/ submodule. scripts/setup.sh, scripts/install_deps.sh, and every project script MUST detect host OS and apply the correct action out of the box (no manual per-OS flags). Compile hardening, jemalloc linkage, and the verification gate apply to both targets.

§109 — Anti-bluff verification gate (load-bearing)

scripts/verify.sh is the single decision point for whether the binary is operator-safe. GREEN → setup.sh proceeds to PATH export; RED → setup.sh REFUSES to install. There is no "force install" mode and no flag that bypasses the gate.

Overrides of the Universal Constitution

None. This project introduces **no override** of any universal clause. Any future override **MUST** appear here under an explicit **### Override §X.Y – <reason>** heading with justification.

Owned-submodule set

Submodules this project owns and is responsible for:

Containers `git@github.com:vasic-digital/Containers.git`
(cgroup orchestration helpers)
constitution `git@github.com:HelixDevelopment/`
HelixConstitution.git (governance; pinned 7f738df)

tmux/ is a **third-party upstream** submodule (tmux/tmux tag 3.6a) — excluded from the owned set; immutable per §107.

Project-specific remotes

Repo	Remotes	Push mechanism
Main — vasic-digital/tmux	github + gitlab	bash <code>commit_all.sh</code> "<msg>" — ONLY allowed mechanism
Containers	github + gitlab	the Containers repo's own commit wrapper

Never `git push` directly on the main repo. SSH remotes only — no HTTPS for Git. No CI/CD pipeline files. Force-push is never automatic and requires explicit per-operation authorization per universal §9.

§11.4.87-98 — universal anchors inherited from constitution 6828ff2 (cycle 2026-05-28)

The constitution submodule was ff-merged from 84c948d to 6828ff2 (2026-05-28). Anchors §11.4.87 through §11.4.98 are universal additions that apply to this project unconditionally per §11.4.35. Long-form bodies live in `constitution/Constitution.md`; short-form mirrors follow.

§11.4.87 — Endless-loop autonomous work + zero-idle agent dispatch + anti-bluff testing

Universal anchor inherited from constitution/Constitution.md §11.4.87 (commit 6828ff2). Operative for this project: agent work continues autonomously until ALL of (a) Issues.md has no In-progress / Ready / In-testing / Reopened items, (b) CONTINUATION.md §3 active-work empty, (c) no in-flight subagent, (d) no external dependency in-flight. Every closed item carries §11.4.4(b) four-layer coverage + captured evidence per §11.4.5/§11.4.69. Tests AND HelixQA Challenges bound equally. **Canonical authority:** constitution submodule Constitution.md §11.4.87 (6828ff2).

§11.4.88 — Background-push: commit-lock release immediately, push detached

Universal anchor inherited from constitution/Constitution.md §11.4.88 (commit 6828ff2). Operative: `commit_all.sh` MUST release its commit flock the moment `git commit` returns 0; `git push` runs detached via `nohup ... & + disown`; per-remote `.git/.push.<remote>.lock` serialises same-remote concurrent pushes but allows different-remote parallelism. Backgrounded push failures land under `qa-results/push_failures/`. The only escape is `--sync-push` for §11.4.41 force-push paths. **Canonical authority:** constitution submodule Constitution.md §11.4.88 (6828ff2).

§11.4.89 — Background test execution (long tests run detached)

Universal anchor inherited from constitution/Constitution.md §11.4.89 (commit 6828ff2). Operative for this project's >30 s tests (`bash scripts/tests/run_all.sh`, `meta_test_false_positive_proof.sh`, `test_e2e.sh`, etc.): MUST spawn via `nohup ... > <log> 2>&1 & + disown`; main work stream returns to the §11.4.42 priority queue; hard-dependent next steps poll exit-status. Per-script flock serialises same-script invocations. **Canonical authority:** constitution submodule Constitution.md §11.4.89 (6828ff2).

§11.4.90 — Obsolete status + per-item obsolescence audit

Universal anchor inherited from constitution/Constitution.md §11.4.90 (commit 6828ff2). The §11.4.15 Status closed-set gains terminal value `Obsolete` (→ `Fixed.md`) (orthogonal to `Type`). Reasons drawn from closed vocabulary: `superseded-by-design-change` |

superseded-by-later-mandate | feature-removed | duplicate-of | unsupported-topology. Every Obsolete heading carries ****Obsolete-Details:**** (Since + Reason + Superseding-item

- Triple-check evidence) within 8 non-blank lines. §11.4.23 colorizer adds cell-status-obsolete class (light-gray + strikethrough). **Canonical authority:** constitution submodule Constitution.md §11.4.90 (6828ff2).

§11.4.91 — Summary-doc clarity mandate

Universal anchor inherited from constitution/Constitution.md §11.4.91 (commit 6828ff2). Every summary one-liner (Issues_Summary.md, Fixed_Summary.md, README doc-link, Status_Summary pages) **MUST** be self-contained — ≥ 6 words OR ≥ 40 chars naming SUBJECT + PROBLEM/GOAL. Forbidden anti-patterns: section labels (Composes with, etc.); bare metadata fragments (Critical, Bug, In progress); $\S$ -letter alone. Generators **MUST** extract from H1/H2 heading line per §11.4.54 ATM-NNN convention and refuse anti-pattern rows. **Canonical authority:** constitution submodule Constitution.md §11.4.91 (6828ff2).

§11.4.92 — Multi-pass change-evaluation discipline

Universal anchor inherited from constitution/Constitution.md §11.4.92 (commit 6828ff2). Every non-trivial change passes 5 evaluation passes **BEFORE** commit-ready: (1) main-task captured-evidence verification; (2) regression blast-radius enumeration of every dependent file; (3) cross-feature interaction analysis (shared state/timing/hardware); (4) deep-research per §11.4.8 + CodeGraph queries per §11.4.78/§11.4.79; (5) anti-bluff confirmation. Per-pass written documentation required. Trivial exemption only for zero-source-touch revision-bump / typo commits citing the exemption. **Canonical authority:** constitution submodule Constitution.md §11.4.92 (6828ff2).

§11.4.93 — SQLite-backed single source of truth for workable items

Universal anchor inherited from constitution/Constitution.md §11.4.93 (commit 6828ff2). Text-based Issues/Fixed/Summary/CONTINUATION constellation converts to SQLite-DB-backed single source of truth at docs/workable_items.db (TRACKED per §11.4.95, not gitignored). A Go binary at cmd/workable-items/ provides bidirectional sync (md-to-db / db-to-md / diff / validate / add / close). Round-trip **MUST** be byte-identical within whitespace-and-section-order tolerance. commit_all.sh refuses to commit while diff is non-empty. **Canonical authority:** constitution submodule Constitution.md §11.4.93 (6828ff2).

§11.4.94 — Zero-idle priority-first parallel-by-default operating mode

Universal anchor inherited from constitution/Constitution.md §11.4.94 (commit 6828ff2). Binds §11.4.20+§11.4.42+§11.4.58+§11.4.70+§11.4.72+§11.4.82+§11.4.87+§11.4.88+§11.4.89 into a single always-on enforcement: idle ONLY when every queued item is genuinely externally blocked OR operator STOP OR §12 host-safety; before any wake/sleep the conductor MUST survey parallel-work feasibility, identify non-contending items, dispatch in parallel per §11.4.20/§11.4.70 subagent-driven default + §11.4.58 PWU disjoint scope + §11.4.89 background long tests. **Canonical authority:** constitution submodule Constitution.md §11.4.94 (6828ff2).

§11.4.95 — Workable-items SQLite DB TRACKED in git, NEVER gitignored

Universal anchor inherited from constitution/Constitution.md §11.4.95 (commit 6828ff2). Amends §11.4.93: docs/workable_items.db IS authoritative source data, NOT a build artefact — it is TRACKED in git, NEVER gitignored. Every workable-items sync md-to-db mutation stages + commits + pushes the DB alongside the MD regen per §11.4.19 atomic-move + §2.1 multi-upstream push. WAL-checkpoint (PRAGMA wal_checkpoint(TRUNCATE)) required before commit-stage. Destructive DB ops require §9.2 hardlinked-backup. **Canonical authority:** constitution submodule Constitution.md §11.4.95 (6828ff2).

§11.4.96 — Safe-parallel-work-with-long-build catalogue + mandate

Universal anchor inherited from constitution/Constitution.md §11.4.96 (commit 6828ff2). Operational catalogue for long-running build workloads classifies parallel work as SAFE (docs/scripts/tests/submodule pushes under §11.4.88 + research + DB ops + subagent dispatch) vs UNSAFE (destructive git ops on source tree, mass deletes under hot subtrees, container destruction, host-safety breaches). Conductor MUST consult catalogue at every pause point and dispatch ≥ 1 safe item per §11.4.20/ §11.4.70 subagent default + §11.4.89 background. **Canonical authority:** constitution submodule Constitution.md §11.4.96 (6828ff2).

§11.4.97 — Maximum-use-of-idle-time + progress-update cadence

Universal anchor inherited from constitution/Constitution.md §11.4.97 (commit 6828ff2). Strengthens §11.4.87+§11.4.94+§11.4.96: every idle minute during which work could autonomously progress and is not genuinely blocked = §11.4.97 violation. Progress-update cadence: emit operator-facing 1-line update at every commit landed / subagent return / constitutional anchor landing / captured-evidence acquisition / milestone closure — no operator prompt required. Continuous physical-proof gathering per §11.4.5+§11.4.6+§11.4.69. **Canonical authority:** constitution submodule Constitution.md §11.4.97 (6828ff2).

§11.4.98 — Full-Automation Anti-Bluff — live tests re-runnable end-to-end without manual intervention

Universal anchor inherited from constitution/Constitution.md §11.4.98 (commit 6828ff2). Closes the manual-intervention gap §11.4.85/§11.4.87/ §11.4.89/§11.4.94 did not explicitly forbid: every test (unit / integration / e2e / Challenge / stress / chaos / live) MUST be fully self-driving end-to-end and report PASS/FAIL/SKIP-with-reason without any further human action after startup. One-time credential bootstrap OUTSIDE test execution is the only permissible exception. Re-runnability proof = PASS at -count=3 consecutive automated invocations with self-cleaning state. Manual-dependency tests not rewritten within 30 days graduate to §11.4.90 Obsolete citing §11.4.98 as the reason. **Canonical authority:** constitution submodule Constitution.md §11.4.98 (6828ff2).

§11.4.99 — Latest-Source Documentation Cross-Reference Mandate

Universal anchor inherited from constitution/Constitution.md §11.4.99 (commit 9e3bcc5 ff 2026-05-28). Before committing any operator-facing instruction / guide / manual / setup walkthrough / troubleshooting cookbook / API how-to, the author MUST cross-reference each instruction against the LATEST official online documentation of the service / library being documented via WebFetch / MCP / direct browsing — NOT training data. A ## Sources verified <date> section in the doc and a Sources verified <date>: <urls> footer in the commit message are MANDATORY. Re-verification cadence: every major-release boundary; ≤ 6 months for general docs; ≤ 90 days for risk-classified service families (messengers, cloud APIs, payments, LLM/AI providers, code-hosting, OS/package managers). Stale docs → §11.4.90 Obsolete after the 30-day grace window. **Canonical authority:**

constitution submodule [Constitution.md §11.4.99 \(9e3bcc5\)](#).
Forensic anchor: Herald MTProto guide near-miss case study
where stale guidance could have caused a Telegram account ban.

§11.4.100 — RETIRED (per §11.4.17/§11.4.35, demoted to consumer project as project-specific)

§11.4.101 — Autonomous-decision-over-blocking mandate (User mandate, 2026-05-28)

Universal anchor inherited from [constitution/Constitution.md §11.4.101 \(commit 6828ff2\)](#). In autonomous/endless-loop mode, the agent MUST minimize operator-blocking and make the safe, reliable, reversible decision autonomously so work is not stalled waiting for input. Decision rule (ALL hold): action is reversible OR has a captured pre-op backup (§9.2); safe choice from captured evidence (§11.4.6 no-guessing); wrong-choice blast radius bounded AND recoverable. Block only when irreversible AND high-blast-radius AND safe choice cannot be determined from evidence. An unavoidable block parks one work unit, not the loop. **Canonical authority:** constitution submodule [Constitution.md §11.4.101 \(commit 1576d3d\)](#).

§11.4.102 — Mandatory systematic-debugging + always-loaded skill-discovery + plugin-dependency availability (User mandate, 2026-05-29)

Universal anchor inherited from [constitution/Constitution.md §11.4.102 \(commit 6828ff2\)](#). On ANY spotted issue/bug/test-failure/gate-failure/regression, the agent MUST activate `superpowers:systematic-debugging` BEFORE proposing/writing/applying any fix — NO FIXES WITHOUT ROOT CAUSE INVESTIGATION FIRST. `superpowers:using-superpowers` MUST be always-loaded and consulted before any task. All skill plugins/dependencies MUST be installed + loadable before dependent work proceeds. **Canonical authority:** constitution submodule [Constitution.md §11.4.102](#).

§11.4.103 — Continuous parallel-stream working routine (User mandate, 2026-05-29)

Universal anchor inherited from [constitution/Constitution.md §11.4.103](#). The main work stream MUST always stay FREE (commit AND push operations run detached). ≥3 parallel background streams at all times with auto-backfill. Audio always

top per §11.4.72. Idle ONLY when genuinely externally blocked.
Canonical authority: constitution submodule [Constitution.md](#) §11.4.103.

§11.4.108 — Four-layer fix-verification + runtime-signature-as-definition-of-done mandate

Universal anchor inherited from constitution/Constitution.md
§11.4.108. A fix crosses FOUR distinct layers and "fixed" at one does NOT imply fixed at the next: SOURCE → ARTIFACT → RUNTIME-ON-CLEAN-TARGET → USER-VISIBLE. Every fix declares ONE machine-checkable runtime signature verified on a clean target. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.108.

§11.4.113 — Absolute no-force-push + merge-onto-latest-main mandate (User mandate, 2026-06-03)

Universal anchor inherited from constitution/Constitution.md
§11.4.113. Force-push is STRICTLY FORBIDDEN with NO exception. The mandated 6-step integration procedure for any repo/submodule whose local has commits to publish OR whose mirrors diverged: fetch → set base → merge → resolve → commit → push. Each push is a fast-forward because the merge commit descends from every mirror tip. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.113. This project installs a PreToolUse guard that blocks --force / --force-with-lease at the tool-call boundary.

§11.4.115 — RED-baseline-on-the-broken-artifact + polarity-switch mandate

Universal anchor inherited from constitution/Constitution.md
§11.4.115. Every RED test MUST reproduce the defect on the CURRENT pre-fix artifact with positive captured evidence. The SAME test source carries RED_MODE=1→0 polarity switch — RED-on-broken then GREEN-on-fixed. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.115.

§11.4.122 — No-silent-removal-of-existing-components-without-operator-confirmation (User mandate, 2026-06-03)

Universal anchor inherited from constitution/Constitution.md
§11.4.122. No application/component/service/package/feature/driver/module/prebuilt — any end-user capability — may be

removed without FIRST interactively asking the operator and receiving an EXPLICIT keep-or-remove decision. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.122.

§11.4.123 — Rock-solid-proof-or-deep-research mandate (User mandate, 2026-06-03)

Universal anchor inherited from constitution/Constitution.md §11.4.123. Every reported issue/fix/claimed completion MUST be 100% validated with rock-solid CAPTURED proof per §11.4.5/§11.4.69/§11.4.107. When unsure how to validate, perform deep web research FIRST — declaring "untestable" without exhausting research is itself a violation. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.123.

§11.4.124 — Dead/unwired-code investigate-before-remove mandate (User mandate, 2026-06-04)

Universal anchor inherited from constitution/Constitution.md §11.4.124. "Zero importers $\Rightarrow$ dead $\Rightarrow$ delete" is a GUESS (§11.4.6). Before removing ANY seemingly-dead element, investigate via git history and capture FACT where/how it was wired in and how it became dead. Removal permitted ONLY with captured proof. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.124.

§11.4.125 — Code-review-agent gate before pre-build + main build (mandatory multi-layer review)

Universal anchor inherited from constitution/Constitution.md §11.4.125. After all batch fixes/changes and BEFORE the pre-build sweep + main build, dispatch dedicated code-review agent(s) analyzing all work done + existing data/facts + codebase + git history. Any finding fixed + re-tested + re-reviewed before build proceeds. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.125.

§11.4.126 — Default autonomous-loop working mode from first prompt (User mandate, 2026-06-04)

Universal anchor inherited from constitution/Constitution.md §11.4.126. The endless fully-autonomous loop is the DEFAULT working mode, engaged automatically from the FIRST prompt. Continues until release tag published OR current scope fully completed AND queue empty. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.126.

§11.4.127 — Session-handoff resumption-prompt mandate (User mandate, 2026-06-06)

Universal anchor inherited from constitution/Constitution.md §11.4.127. On fresh-session need OR operator ask, the agent MUST prepare + provide a ready-to-paste resumption prompt valid for that EXACT moment — self-contained, pointing to handoff docs, stating current PHASE/NEXT/terminal-goal, embedding exact live-state anchors. **Canonical authority:** constitution submodule Constitution.md §11.4.127.

§11.4.131 — Standing session-resumption file mandate (User mandate, 2026-06-07)

Universal anchor inherited from constitution/Constitution.md §11.4.131. Every project MUST maintain a SINGLE canonical always-current session-resumption file — the OUT-OF-THE-BOX entry point. (Re)written whenever fresh session needed OR live state materially changes. Points to handoff docs, embeds live-state anchors. **Canonical authority:** constitution submodule Constitution.md §11.4.131.

§11.4.132 — Risk-ordered validation priority mandate (User mandate, 2026-06-07)

Universal anchor inherited from constitution/Constitution.md §11.4.132. Tests/validations MUST run in RISK-DESCENDING order: most-recently-worked → historically-most-problematic → highest-crash-likelihood → most-reopened. Each highest-risk item GREEN with captured proof before the rest runs. **Canonical authority:** constitution submodule Constitution.md §11.4.132.

§11.4.134 — Code-review iterate-until-GO + rock-solid-evidence mandate (User mandate, 2026-06-08)

Universal anchor inherited from constitution/Constitution.md §11.4.134. When code-review returns ANY finding, the review MUST BE RE-RUN after each remediation until a clean GO with ZERO new issues AND ZERO warnings. Every round carries rock-solid physical evidence. **Canonical authority:** constitution submodule Constitution.md §11.4.134.

§11.4.135 — Standing regression-guard suite + every-fixed-defect-gets-a-permanent-regression-test (User mandate, 2026-06-08)

Universal anchor inherited from constitution/Constitution.md
§11.4.135. Every project MUST maintain a standing regression-guard suite. Every closed defect MUST register a §11.4.115 RED-on-broken-artifact regression test. The suite runs FIRST in every post-deploy cycle and BLOCKS release tag on failure. **Canonical authority:** constitution submodule Constitution.md §11.4.135.

§11.4.138 — Operator-escape ⇒ mandatory bluff-audit + permanent guard (User mandate, 2026-06-08)

Universal anchor inherited from constitution/Constitution.md
§11.4.138. When the operator finds a defect the GREEN test suite passed, this is a §11.4 PASS-bluff — triggers systematic-debugging, a bluff-audit citing the exact assertion that should have caught it, a permanent regression guard, and the audit committed under docs/research/. **Canonical authority:** constitution submodule Constitution.md §11.4.138.

§11.4.139 — Fresh-process clean-artifact runtime-signature mandate (User mandate, 2026-06-08)

Universal anchor inherited from constitution/Constitution.md
§11.4.139. Before any post-deploy validation, the harness MUST assert running-artifact == built-artifact — deployment yields a CLEAN target OR pre-validation proves no stale overlay shadows the deployed code. **Canonical authority:** constitution submodule Constitution.md §11.4.139.

§11.4.140 — Universal action-prefix system (User mandate, 2026-06-09)

Universal anchor inherited from constitution/Constitution.md
§11.4.140. When a user prompt's FIRST non-blank line starts with a recognised action token (e.g. BACKGROUND :: <rest>), look up the action registry and REPLACE the prefix with the action's expansion text. Four equivalent forms. Unknown tokens matching the grammar are NEVER silently expanded — ask or treat literally. **Canonical authority:** constitution submodule Constitution.md §11.4.140.

§11.4.141 — Token-efficiency mandate (User mandate, 2026-06-09)

Universal anchor inherited from constitution/Constitution.md
§11.4.141. Every AI-coding-agent project MUST cut token spend 60–70% via: prompt-cache governance prefix, subagent model-tiering (Haiku for mechanical work), thin always-loaded index + on-demand detail, CodeGraph-first navigation, output-token reduction, tool-call batching. Measured proof with zero regression required. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.141.

§11.4.142 — Universal code-review mandate — every change reviewed, always, no exception (User mandate, 2026-06-09)

Universal anchor inherited from constitution/Constitution.md
§11.4.142. EVERY change — without exception — MUST pass through an INDEPENDENT code-review step BEFORE acceptance/commit/build. No "trivial change" or "doc edit" exemption. Reviewer structurally separate from author. Iterates to clean GO per §11.4.134. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.142.

§11.4.143 — Real-user-journey mandate for video-streaming-app full-automation tests (User mandate, 2026-06-10)

Universal anchor inherited from constitution/Constitution.md
§11.4.143. Every full-automation test asserting a video player / streaming app plays content MUST drive the REAL end-user journey — launch → catalog browse → specific-title selection → real Play/Resume press → confirm chosen content plays with correct subtitles. No shortcuts/samples/deep-links. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.143.

§11.4.144 — Tracked/recorded-device availability-following mandate (User mandate, 2026-06-10)

Universal anchor inherited from constitution/Constitution.md
§11.4.144. Every tracked/recorded device MUST be availability-FOLLOWED — connection state monitored, any drop logged as an honest offline event → wait → re-attach → escalate. A silent corpus hole is a §11.4 PASS-bluff. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.144.

§11.4.145 — Independent multi-angle impact-research per change (User mandate, 2026-06-10)

Universal anchor inherited from constitution/Constitution.md
§11.4.145. For EVERY change, INDEPENDENT impact-research agents MUST research across 8 CLOSED-SET angles: correctness/logic, regression, latent/dangerous-code, security, performance, host/data/target-hardware safety, cross-feature interaction, business-logic conformance. Output = GO/NO-GO; iterates to clean GO. **Canonical authority:** constitution submodule Constitution.md §11.4.145.

§11.4.146 — Reproduce-first test + same-test-confirms-fix + mandatory extend-to-all-cases workflow (User mandate, 2026-06-10)

Universal anchor inherited from constitution/Constitution.md
§11.4.146. Three-step workflow: (STEP 1) RED test as §11.4.115 reproduce-first + forensic characterisation; (STEP 2) SAME test RED_MODE=1→0 GREEN confirmation; (STEP 3) fan-out across full case-space with enumerated coverage. **Canonical authority:** constitution submodule Constitution.md §11.4.146.

§11.4.147 — Crashed-agent respawn-until-complete + no-work-loss registry mandate (User mandate, 2026-06-10)

Universal anchor inherited from constitution/Constitution.md
§11.4.147. Every dispatched agent MUST be tracked through its full lifecycle (durable registry with CLOSED SET statuses: dispatched/in-flight/crashed/respawned/complete). Crash ≠ done — mechanical respawn until complete. Partial state preserved → quiescence-checked → resume-or-clean-restart. **Canonical authority:** constitution submodule Constitution.md §11.4.147.

§11.4.148 — Workable-item integrity (status+type+id) + comprehensive structured description + bidirectional external-tracker sync (User mandate, 2026-06-10)

Universal anchor inherited from constitution/Constitution.md
§11.4.148. No item without valid status+type+stable id on ALL surfaces. Comprehensive structured description mandatory. BLOCKED items carry WHY + enumerated unblock choices. Regular bidirectional DB↔docs↔tracker sync via idempotent push. **Canonical authority:** constitution submodule Constitution.md §11.4.148.

§11.4.149 — Per-workable-item testing-diary mandate (User mandate, 2026-06-10)

Universal anchor inherited from constitution/Constitution.md
§11.4.149. Every workable item carries an append-only TESTING DIARY (date_time, tested_by, result, observations, evidence_path). PASS requires non-empty evidence path (schema constraint). Four-format exports + tracker SUB-TASK model. Minimal-LLM deterministic tooling. **Canonical authority:** constitution submodule Constitution.md §11.4.149.

§11.4.150 — Mandatory deep multi-angle web research per change/issue (User mandate, 2026-06-11)

Universal anchor inherited from constitution/Constitution.md
§11.4.150. For EVERY fix/improvement/closure, perform a documented deep multi-angle web research pass (≥ 2 distinct angles). No closure-as-fixed/structural WITHOUT documented pass. Latest-source cited. Run in parallel with main stream. **Canonical authority:** constitution submodule Constitution.md §11.4.150.

§11.4.151 — Project-prefixed release-tag/version-naming mandate (User mandate, 2026-06-12)

Universal anchor inherited from constitution/Constitution.md
§11.4.151. Every release tag and version name MUST be prefixed <PREFIX>-<version>. Prefix resolution: HELIX_RELEASE_PREFIX from .env (authoritative) else lowercased project-root dir name. SAME prefix across main repo + all owned submodules in one release. **Canonical authority:** constitution submodule Constitution.md §11.4.151. (Note: this project's current tag history uses vN.N.N without prefix — operator decision to adopt prefix deferred to a future major release.)

§11.4.152 — Crashlytics-recorded-data continuous monitoring + systematic-debug + regression-test-coverage mandate (User mandate, 2026-06-13)

Universal anchor inherited from constitution/Constitution.md
§11.4.152. Every project with Firebase Crashlytics enabled MUST continuously monitor ALL four surfaces (fatal crashes, ANRs, performance traces, non-fatals), systematic-debug each, and cover every closed issue with a permanent §11.4.135 regression guard. **Canonical authority:** constitution submodule Constitution.md §11.4.152.

§11.4.153 — Comprehensive per-feature Status + Status_Summary document set with mandatory video-recording confirmation (User mandate, 2026-06-15)

Universal anchor inherited from constitution/Constitution.md
§11.4.153. Every project MUST maintain under docs/features/ a comprehensive feature Status set enumerating EVERY component/app/surface/feature. Per-feature fields include Video-recording confirmation (path to real-use video). Four-format export (HTML+PDF+DOCX). **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.153.

§11.4.154 — Window-scoped capture + fresh-corpus rotation for feature/QA recordings (User mandate, 2026-06-15)

Universal anchor inherited from constitution/Constitution.md
§11.4.154. Every video MUST capture ONLY the target app/service window, NEVER the whole desktop. Fresh-corpus rotation: remove agent's own prior in-scope stale recordings before new recording begins. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.154.

§11.4.155 — Project-name-prefixed feature/QA recording filenames (User mandate, 2026-06-15)

Universal anchor inherited from constitution/Constitution.md
§11.4.155. Every recording filename MUST start with the project-name prefix (from HELIX_RELEASE_PREFIX or lowercased project-root dir). Format: <PREFIX>---<feature>---<run-id>.<ext>. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.155.

§11.4.156 — All CI/CD automation MUST be disabled (User mandate, 2026-06-15)

Universal anchor inherited from constitution/Constitution.md
§11.4.156. All governed repos MUST ship with ALL server-side CI/CD automation DISABLED — no live .github/workflows/*.yaml or .gitlab-ci.yaml in any authored repo. Enforcement via local git hooks + pre-tag sweep. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.156.

§11.4.157 — GEMINI.md maintained in lockstep with CLAUDE.md / AGENTS.md / QWEN.md (User mandate, 2026-06-15)

Universal anchor inherited from constitution/Constitution.md
§11.4.157. GEMINI.md is a FIRST-CLASS governance carrier EQUAL to CLAUDE.md/AGENTS.md/QWEN.md — no governance change is complete until GEMINI.md carries it alongside the other three mirrors. No silent drift. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.157.

§11.4.158 — Intensive all-feature/flow/edge-case video-recording + read-the-screen content-verification mandate (User mandate, 2026-06-16)

Universal anchor inherited from constitution/Constitution.md
§11.4.158. Every project MUST be covered by intensive automated testing that exercises + RECORDS every feature/flow/use-case/edge-case. The System MUST ACTUALLY READ every shown log line/message and VERIFY it is a genuine working result. HelixQA drives the exercise→record→read→score pass. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.158.

§11.4.159 — Mandatory window-specific video recording + vision validation mandate (User mandate, 2026-06-20)

Universal anchor inherited from constitution/Constitution.md
§11.4.159. Every feature test/validation/QA session producing video MUST: window-specific capture only, MP4 format (H.264), project-name prefixed, mandatory vision validation after recording with content-verification workflow (SPECIFY→RECORD→EXTRACT→VERIFY→ACCEPT/REJECT). **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.159.

§11.4.160 — Vision-verified recording + HelixQA bridge mandate (User mandate, 2026-06-21)

Universal anchor inherited from constitution/Constitution.md
§11.4.160. Every video recording MUST be processed through a vision/OCR pipeline against SPECIFY-phase patterns before acceptance. Bridge to HelixQA for automated verification (≤5s frame interval, self-validated analyzer). **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.160.

§11.4.161 — Rootless container runtime mandate (User mandate, 2026-06-21)

Universal anchor inherited from constitution/Constitution.md
§11.4.161. Every project MUST use Podman in rootless mode for ALL containerized workloads. The vasic-digital/containers Submodule is the sole container orchestration layer; extend upstream, never reimplement. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.161. (This project's Containers/ submodule already complies.)

§11.4.162 — OpenDesign UI design system mandate (User mandate, 2026-06-21)

Universal anchor inherited from constitution/Constitution.md
§11.4.162. Every project producing user-facing interfaces MUST use OpenDesign as the mandatory UI design-and-refinement system — design tokens for color (light+dark), typography, spacing, component-level tokens. Extend upstream per §11.4.74. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.162.

§11.4.163 — Universal Media Validation & Verification Mandate (User mandate, 2026-06-21)

Universal anchor inherited from constitution/Constitution.md
§11.4.163. Every recorded artifact MUST pass a MEDIA VALIDATION pipeline before acceptance — OCR/transcription/text parsing vs expected patterns, self-validated golden-good/golden-bad analyzer. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.163.

§11.4.164 — Universal Constitution Auto-Propagation & Hook System (User mandate, 2026-06-21)

Universal anchor inherited from constitution/Constitution.md
§11.4.164. Every constitution fetch+pull MUST trigger constitution/scripts/post_update_hook.sh (inherited by reference, NEVER copied) that detects changed files, registers new skills/ MCP/hooks/scripts, validates syntax. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.164.

§11.4.165 — Universal Independent Verification Agent Mandate (User mandate, 2026-06-21)

Universal anchor inherited from constitution/Constitution.md
§11.4.165. Every code change OR recorded media artifact MUST pass an INDEPENDENT verifier structurally separate from the author, producing structured findings with evidence paths, iterating to GO per §11.4.134. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.165.

§11.4.166 — REPEALED (operator decision, 2026-06-22)

The Universal Semgrep static-analysis mandate is repealed — Semgrep no longer mandatory. Anchor 11.4.166 retired.

§11.4.167 — Big-work-item feature work-stream lifecycle (User mandate, 2026-06-23)

Universal anchor inherited from constitution/Constitution.md
§11.4.167. Every BIG work item MUST be developed as its own isolated feature work-stream — CoW/reflink sibling copy, own feature/<slug> branch, own per-feature builds + tags, kept separate from trunk until operator approves. Trunk merged INTO every live stream frequently. **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.167.

§11.4.168 — Exported-document independent content + textual + full-visual validation mandate (User mandate, 2026-06-23)

Universal anchor inherited from constitution/Constitution.md
§11.4.168. Every exported document MUST pass INDEPENDENT validation across THREE layers: (1) CONTENT — faithful, (2) TEXTUAL — no raw diagram-source as body text, (3) FULL VISUAL — embedded diagrams render as images. Self-validated analyzer per §11.4.107(10). **Canonical authority:** constitution submodule [Constitution.md](#) §11.4.168.

§11.4.169 — Mandatory comprehensive test-type coverage with anti-bluff captured evidence (User mandate, 2026-06-25)

Universal anchor inherited from constitution/Constitution.md
§11.4.169. Every project MUST have this CLOSED enumerated test-type set aiming 100% coverage: unit, integration, e2e, full-automation, Challenges, HelixQA, DDoS/load-flood, security, stress+chaos, concurrency/atomicity, race-condition/deadlock,

memory, benchmarking/performance. Each PASS cites §11.4.5/.69/.107 physical evidence. **Canonical authority:** constitution submodule Constitution.md §11.4.169.

§11.4.170 — Device-independent host-side rendered-UI visual-proof mandate (User mandate, 2026-06-25)

Universal anchor inherited from constitution/Constitution.md §11.4.170. Every UI change MUST be proven by host-side rendered pixels (Compose→Roborazzi/Paparazzi, web→Playwright, SwiftUI→snapshot) for EVERY screen×state×{light,dark} theme. VALUE/token-equality UI tests FORBIDDEN as proof. Self-validated analyzer. "Device offline" is NEVER a valid skip. **Canonical authority:** constitution submodule Constitution.md §11.4.170.

This Project Constitution governs the vasic-digital/tmux repository. It is subordinate to and extends constitution/Constitution.md.